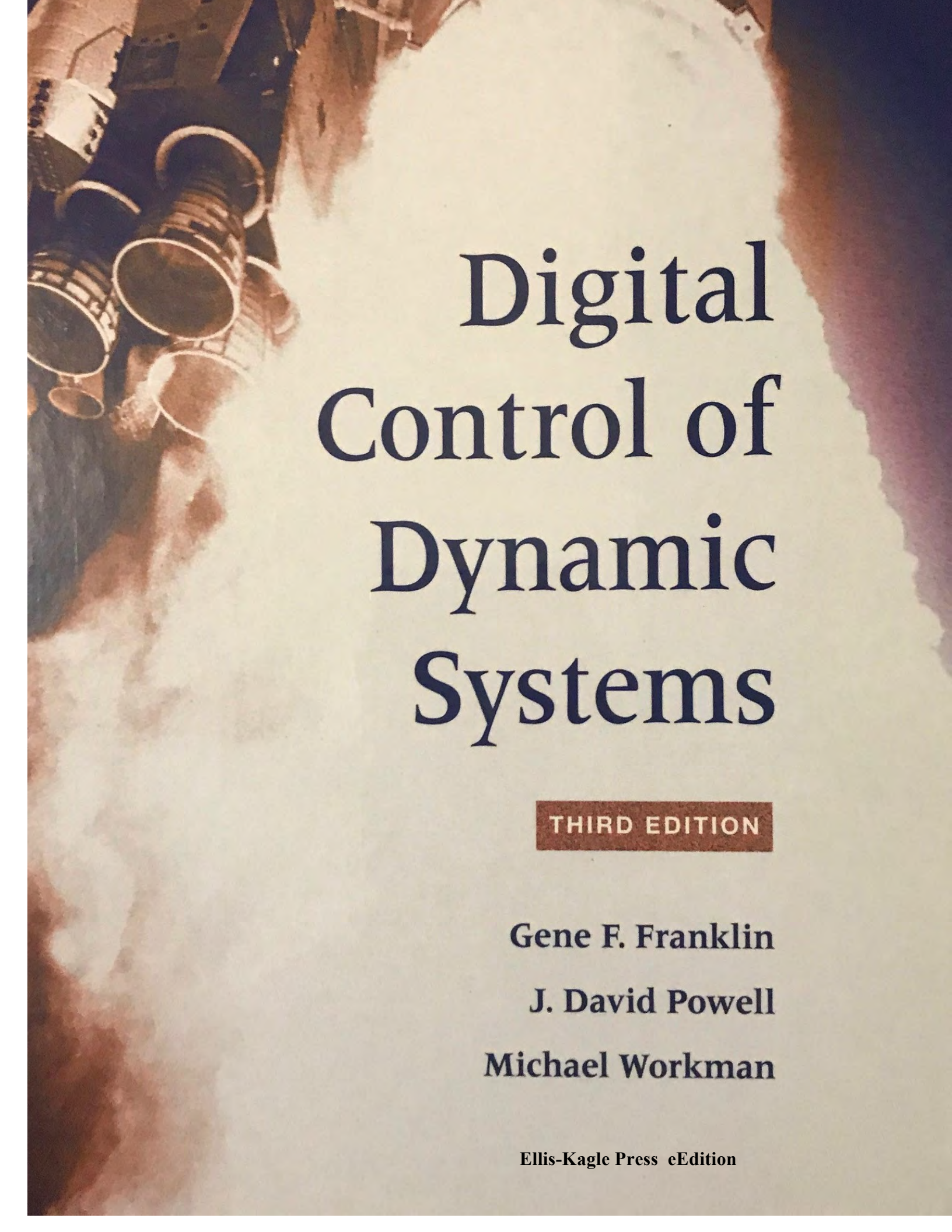


The background of the cover is a photograph of a space shuttle launch. The shuttle is on the left, angled upwards, with its three main engines visible. A large, billowing plume of white smoke and orange-brown fire surrounds the shuttle, filling most of the frame. The sky is a deep blue at the top right.

# Digital Control of Dynamic Systems

THIRD EDITION

Gene F. Franklin

J. David Powell

Michael Workman

Ellis-Kagle Press eEdition

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This well-respected, market-leading text discusses the use of digital computers in the real-time control of dynamic systems. The emphasis is on the design of digital controls that achieve good dynamic response and small errors while using signals that are sampled in time and quantized in amplitude. Both transform-based and state-based classical and modern control methods are described and applied to illustrative examples. The strengths and limitations of each method are explored to help the reader develop solid designs with the least effort.

The third edition includes two chapters which offer a review of feedback control systems and an overview of digital control. The eEdition version is the same, except that chapters 13 and 14 are not included. They can be obtained by sending a request to [EllisKagle@gmail.com](mailto:EllisKagle@gmail.com).

MATLAB is thoroughly integrated throughout the text, in exposition and in problems, to offer readers a complete design picture. The 3rd edition has been updated to be fully compatible with MATLAB versions 4 and 5 and includes twice as many end-of-chapter problems as the previous edition.

**Gene F. Franklin** was Professor Emeritus of Electrical Engineering at Stanford University where he served as Vice Chairman and Acting Chair of the department. In addition to directing research in feedback control, he has served as a consultant to several computer manufacturers on digital control of position mechanisms. He was a life Fellow of IEEE and delivered the IEEE Bode Lecture for 1994. He has been recognized for his work in the classroom with the American Automatic Control Council's Education Award. Sadly, he passed away in August, 2012.

**J. David Powell** is Professor(Emeritus), of Aeronautics and Astronautics and Mechanical Engineering at Stanford University. Professor Powell's most recent research focused on the use of GPS for air and ground navigation. In addition, he served as a consultant to the automotive industry, heavy equipment manufacturers, and aerospace companies on a wide range of controls problems . He is a Fellow of the AIAA and ASME.

**Michael L. Workman** was Vice President of Development Server/Subsystem Products for IBM's Storage Products Company. He was responsible for the worldwide development, launch, and support of SSD Hard Disk Drives for Server Product Families development. He then became the CEO of Pillar Data Systems, which became part of the Oracle Corporation.

Also by **Gene Franklin and Dave Powell (with Abbas Emami-Naeini)**,  
**Feedback Control of Dynamics Systems**, Eighth Edition (2019), Pearson Education, Inc.

THIRD EDITION

# ♦ Digital Control ♦ of Dynamic Systems

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Gene F. Franklin  
Stanford University

J. David Powell  
Stanford University

Michael L. Workman  
Oracle Corporation



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1200 Pilarcitos Ave.  
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elliskagle@gmail.com  
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### **Tribute to Gene Franklin**

It is with great personal sadness that we report the passing of Prof. Gene Franklin on August 9, 2012. Gene was a mentor, teacher and advisor to us. We had many meetings as we collaborated on the writing and publication of this textbook's editions over 35 years, and every single one of those meetings was friendly and enjoyable. We each have expressed different viewpoints over the years on various topics, but we were always able to encompass the views into the book in a friendly and collaborative manner. We learned control along with humor from Gene in grad school classes, and we benefited from his mentoring; in one case as a new Assistant Professor, and in the other as a PhD advisee. Gene always had a smile with a twinkle in his eye and was a pleasure to work with. We have lost a dear friend and colleague. Gene was a true gentleman.

J.D.P.  
M.W.



*To Valerie, Daisy, Annika,  
Davenport, Patti, Lauren,  
Max, and to the memory of  
Lorna and Gene*



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# ♦ Preface ♦

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This book is about the use of digital computers in the real-time control of dynamic systems such as servomechanisms, chemical processes, and vehicles that move over water, land, air, or space. The material requires some understanding of the Laplace transform and assumes that the reader has studied linear feedback controls. The special topics of discrete and sampled-data system analysis are introduced, and considerable emphasis is given to the  $z$ -transform and the close connections between the  $z$ -transform and the Laplace transform.

The book's emphasis is on designing digital controls to achieve good dynamic response and small errors while using signals that are sampled in time and quantized in amplitude. Both transform (classical control) and state-space (modern control) methods are described and applied to illustrative examples. The transform methods emphasized are the root-locus method of Evans and frequency response. The root-locus method can be used virtually unchanged for the discrete case; however, Bode's frequency response methods require modification for use with discrete systems. The state-space methods developed are the technique of pole assignment augmented by an estimator (observer) and optimal quadratic-loss control. The optimal control problems use the steady-state constant-gain solution; the results of the separation theorem in the presence of noise are stated but not proved.

Each of these design methods—classical and modern alike—has advantages and disadvantages, strengths and limitations. It is our philosophy that a designer must understand all of them to develop a satisfactory design with the least effort.

Closely related to the mainstream of ideas for designing linear systems that result in satisfactory dynamic response are the issues of sample-rate selection, model identification, and consideration of nonlinear phenomena. Sample-rate selection is discussed in the context of evaluating the increase in a least-squares performance measure as the sample rate is reduced. The topic of model making is treated as measurement of frequency response, as well as least-squares parameter estimation. Finally, every designer should be aware that all models are nonlinear

and be familiar with the concepts of the describing functions of nonlinear systems, methods of studying stability of nonlinear systems, and the basic concepts of nonlinear design.

Material that may be new to the student is the treatment of signals which are discrete in time and amplitude and which must coexist with those that are continuous in both dimensions. The philosophy of presentation is that new material should be closely related to material already familiar, and yet, by the end, indicate a direction toward wider horizons. This approach leads us, for example, to relate the  $z$ -transform to the Laplace transform and to describe the implications of poles and zeros in the  $z$ -plane to the known meanings attached to poles and zeros in the  $s$ -plane. Also, in developing the design methods, we relate the digital control design methods to those of continuous systems. For more sophisticated methods, we present the elementary parts of quadratic-loss Gaussian design with minimal proofs to give some idea of how this powerful method is used and to motivate further study of its theory.

The use of computer-aided design (CAD) is universal for practicing engineers in this field, as in most other fields. We have recognized this fact and provided guidance to the reader so that learning the controls analysis material can be integrated with learning how to compute the answers with MATLAB, the most widely used CAD software package in universities. In many cases, especially in the earlier chapters, actual MATLAB scripts are included in the text to explain how to carry out a calculation. In other cases, the MATLAB routine is simply named for reference. All the routines given are tabulated in Appendix E for easy reference; therefore, this book can be used as a reference for learning how to use MATLAB in control calculations as well as for control systems analysis. In short, we have tried to describe the entire process, from learning the concepts to computing the desired results. But we hasten to add that it is mandatory that the student retain the ability to compute simple answers by hand so that the computer's reasonableness can be judged. The First Law of Computers for engineers remains "Garbage In, Garbage Out."

Most of the graphical figures in this third edition were generated using MATLAB<sup>®</sup> supplied by The Mathworks, Inc. The files that created the figures are available from Ellis-Kagle Press at [www.elliskaglepress.com](http://www.elliskaglepress.com). The reader is encouraged to use these MATLAB figure files as an additional guide in learning how to perform the various calculations.

To review the chapters briefly: Chapter 1 contains introductory comments. Chapters 2 and 3 are new to the third edition. Chapter 2 is a review of the prerequisite continuous control; Chapter 3 introduces the key effects of sampling in order to elucidate many of the topics that follow. Methods of linear analysis are presented in Chapters 4 through 6. Chapter 4 presents the  $z$ -transform. Chapter 5 introduces combined discrete and continuous systems, the sampling theorem, and the phenomenon of aliasing.

Chapter 6 shows methods by which to generate discrete equations that will approximate continuous dynamics. The basic deterministic design methods are presented in Chapters 7 and 8—the root-locus and frequency response methods in Chapter 7 and pole placement and estimators in Chapter 8. The state-space material assumes no previous acquaintance with the phase plane or state space, and the necessary analysis is developed from the ground up. Some familiarity with simultaneous linear equations and matrix notation is expected, and a few unusual or more advanced topics such as eigenvalues, eigenvectors, and the Cayley-Hamilton theorem are presented in Appendix C. Chapter 9 introduces optimal quadratic-loss control: First the control by state feedback is presented and then the estimation of the state in the presence of system and measurement noise is developed, based on a recursive least-squares estimation derivation.

In Chapter 10 the nonlinear phenomenon of amplitude quantization and its effects on system error and system dynamic response are studied. Chapter 11 presents methods of analysis and design guidelines for the selection of the sampling period in a digital control system. It utilizes the design methods discussed in Chapters 7, 8, and 9, in examples illustrating the effects of sample rate. Chapter 12 introduces both nonparametric and parametric identification. Nonparametric methods are based on spectral estimation. Parametric methods are introduced by starting with deterministic least squares, introducing random errors, and completing the solution with an algorithm for maximum likelihood. Sub-space methods are also introduced for estimating the state matrices directly. Nonlinear control is the subject of Chapter 13, including examples of plant nonlinearities and methods for the analysis and design of controllers for nonlinear models. Simulation, stability analysis, and performance enhancement by nonlinear controllers and by adaptive designs are also included in Chapter 13. The chapter ends with a nonlinear design optimization alternative to the techniques presented in Chapter 9. The final chapter, 14, is a detailed design example of a digital servo for a disk drive head. Chapters 13 and 14 are not included in this ePUB edition; however, they are available by request to [elliskagle@gmail.com](mailto:elliskagle@gmail.com).

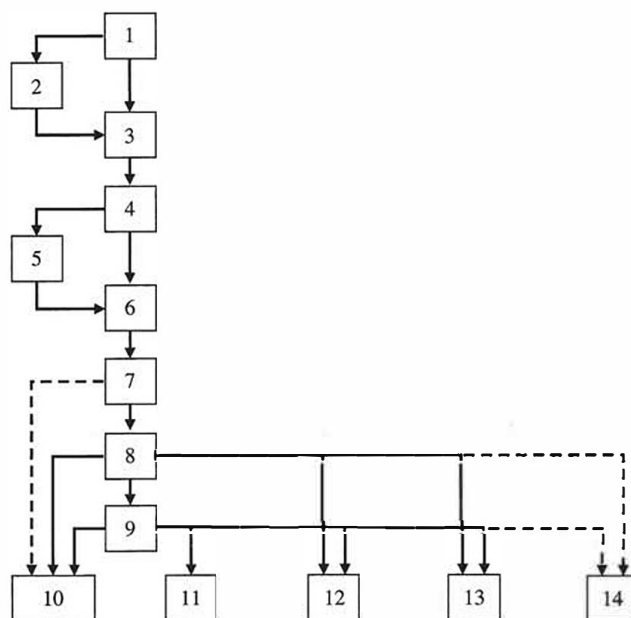
For purposes of organizing a course, Fig. P.1 shows the dependence of material in each chapter on previous chapters. By following the solid lines, the reader will have all the background required to understand the material in a particular chapter, even if the path omits some chapters. Furthermore, sections with a star(\*) are optional and may be skipped with no loss of continuity. Chapters may also be skipped, as suggested by the dashed lines, if the reader is willing to take some details on faith; however, the basic ideas of the later chapters will be understood along these paths.

The first seven chapters (skipping or quickly reviewing Chapter 2) constitute a comfortable one-quarter course that would follow a course in continuous linear control using a text such as Franklin, Powell, and Emami-Naeini (2019). For a one-semester course, the first eight chapters represent a comfortable load. The

Table P.1 Comparison of the Table of Contents

| Chapter Title                         | 3rd Edition<br>Chapter<br>Number | 3rd eEdition<br>Chapter<br>Number |
|---------------------------------------|----------------------------------|-----------------------------------|
| Introduction                          | 1                                | 1                                 |
| Review of Continuous Control          | 2                                | 2                                 |
| Introductory Digital Control          | 3                                | 3                                 |
| Discrete Analysis and the z-Transform | 4                                | 4                                 |
| Sampled Data Systems                  | 5                                | 5                                 |
| Discrete Equivalents                  | 6                                | 6                                 |
| Design Using Transform Methods        | 7                                | 7                                 |
| Design Using State-Space Methods      | 8                                | 8                                 |
| Multivariable and Optimal Control     | 9                                | 9                                 |
| Quantization Effects                  | 10                               | 10                                |
| Sample-Rate Selection                 | 11                               | 11                                |
| System Identification                 | 12                               | 12                                |
| Nonlinear Control                     | 13                               | -                                 |
| Application of Digital Control        | 14                               | -                                 |

Figure P.1



content of a second course has many possibilities. One possibility is to combine Chapters 8 and 9 with Chapter 10, 11, or 12. As can be seen from the figure, many options exist for including the material in the last five chapters. For a full-year course, all fourteen chapters can be covered. One of the changes made in

this third edition is that the optimal control material no longer depends on the least-squares development in the system identification chapter, thus allowing for more flexibility in the sequence of teaching.

It has been found at Stanford that it is very useful to supplement the lectures with laboratory work to enhance learning. A very satisfactory complement of laboratory equipment is a digital computer having an A/D and a D/A converter, an analog computer (or equivalent) with ten operational amplifiers, a digital storage scope, and a CAD package capable of performing the basic computations and plotting graphs as described in [Franklin, Powell (1989)].

There are many important topics in control that we have not been able to include in this book. There is, for example, no discussion of  $\mu$  analysis or design, linear matrix inequalities, or convex optimization. It is our expectation, however, that careful study of this book will provide the student engineer with a sound basis for design of sampled-data controls and a foundation for the study of these and many other advanced topics in this most exciting field.

We wish to acknowledge the vast array of contributors on whose work our own presentation is based. The list of references gives some indication of those to whom we are in debt. On a more personal level, we wish to express our appreciation to Profs. S. Boyd, A. Bryson, R. Cannon, S. Citron, J. How, and S. Rock for their valuable suggestions for the book and especially to our long-time colleague, Prof. Dan DeBra, for his careful reading and many spirited suggestions. We also wish to express our appreciation for many valuable suggestions to the current and former students of E207 and E208, for whom this book was written.

In addition, we want to thank the following people for their helpful reviews of the manuscript: Fred Bailey, University of Minnesota; John Fleming, Texas A&M University; J.B. Pearson, Rice University; William Perkins, University of Illinois; James Carroll, Clarkson University; Walter Higgins, Jr., Arizona State University; Stanley Johnson, Lehigh University; Thomas Kurfess, Georgia Institute of Technology; Stephen Phillips, Case Western Reserve University; Chris Rahn, Clemson University; T. Srinivasan, Wilkes University; Hal Tharp, University of Arizona; Russell Trahan, Jr., University of New Orleans; and Gary Young, Oklahoma State University. We also wish to express our appreciation to Tom Robbins for his help with all the editions of the book, and to Laura Cheu, Emilie Bauer, and all the staff at Addison-Wesley for their quality production of this 3rd edition.

*Stanford, California*

*G.F.F.  
J.D.P.  
M.L.W.*



# ♦ 1 ♦

## Introduction

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### A Perspective on Digital Control

The control of physical systems with a digital computer or microcontroller is becoming more and more common. Examples of electromechanical servomechanisms exist in aircraft, automobiles, mass-transit vehicles, oil refineries, and paper-making machines. Furthermore, many new digital control applications are being stimulated by microprocessor technology including control of various aspects of automobiles and household appliances. Among the advantages of digital approaches for control are the increased flexibility of the control programs and the decision-making or artificial intelligence capability of digital systems, which can be combined with the dynamic control function to meet other system requirements. In addition, one hardware design can be used with many different software variations on a broad range of products, thus simplifying and reducing the design time.

### Chapter Overview

In Section 1.1, you will learn about what a digital control system is, what the typical structure is, and what the basic elements are. The key issues are discussed and an overview of where those issues are discussed in the book is given. Section 1.2 discusses the design approaches used for digital control systems and provides an overview of where the different design approaches appear in the book. Computer Aided Control System Design (CACSD) issues and how the book's authors have chosen to handle those issues are discussed in Section 1.3.

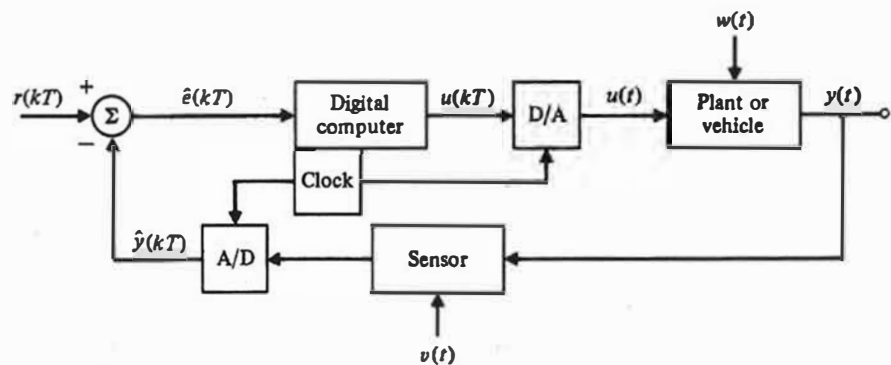
### 1.1 Problem Definition

The digital controls studied in this book are for closed-loop (feedback) systems in which the dynamic response of the process being controlled is a major consideration in the design. A typical structure of the elementary type of system

that will occupy most of our attention is sketched schematically in Fig. 1.1. This figure will help to define our basic notation and to introduce several features that distinguish digital controls from those implemented with analog devices. The process to be controlled (sometimes referred to as the **plant**) may be any of the physical processes mentioned above whose satisfactory response requires control action.

By "satisfactory response" we mean that the plant output,  $y(t)$ , is to be forced to follow or track the reference input,  $r(t)$ , despite the presence of disturbance inputs to the plant [ $w(t)$  in Fig. 1.1] and despite errors in the sensor [ $v(t)$  in Fig. 1.1]. It is also essential that the tracking succeed even if the dynamics of the plant should change somewhat during the operation. The process of holding  $y(t)$  close to  $r(t)$ , including the case where  $r \equiv 0$ , is referred to generally as the process of **regulation**. A system that has good regulation in the presence of disturbance signals is said to have good **disturbance rejection**. A system that has good regulation in the face of changes in the plant parameters is said to have low **sensitivity** to these parameters. A system that has both good disturbance rejection and low sensitivity we call **robust**.

**Figure 1.1**  
Block diagram of a basic  
digital control system



**Notation:**

- $r$  = reference or command inputs
- $u$  = control or actuator input signal
- $y$  = controlled or output signal
- $\hat{y}$  = instrument or sensor output, usually an approximation to or estimate of  $y$ . (For any variable, say  $\theta$ , the notation  $\hat{\theta}$  is now commonly taken from statistics to mean an estimate of  $\theta$ .)
- $e$  =  $r - \hat{y}$  = indicated error
- $e$  =  $r - y$  = system error
- $w$  = disturbance input to the plant
- $v$  = disturbance or noise in the sensor
- A/D = analog-to-digital converter
- D/A = digital-to-analog converter

The means by which robust regulation is to be accomplished is through the control inputs to the plant [ $u(t)$  in Fig. 1.1]. It was discovered long ago<sup>1</sup> that a scheme of feedback wherein the plant output is measured (or sensed) and compared directly with the reference input has many advantages in the effort to design robust controls over systems that do not use such feedback. Much of our effort in later parts of this book will be devoted to illustrating this discovery and demonstrating how to exploit the advantages of feedback. However, the problem of control as discussed thus far is in no way restricted to digital control. For that we must consider the unique features of Fig. 1.1 introduced by the use of a digital device to generate the control action.

We consider first the action of the analog-to-digital (A/D) converter on a signal. This device acts on a physical variable, most commonly an electrical voltage, and converts it into a stream of numbers. In Fig. 1.1, the A/D converter acts on the sensor output and supplies numbers to the digital computer. It is common for the sensor output,  $\hat{y}$ , to be sampled and to have the error formed in the computer. We need to know the times at which these numbers arrive if we are to analyze the dynamics of this system.

sample period

In this book we will make the assumption that all the numbers arrive with the same fixed period  $T$ , called the **sample period**. In practice, digital control systems sometimes have varying sample periods and/or different periods in different feedback paths. Usually there is a clock as part of the computer logic which supplies a pulse or **interrupt** every  $T$  seconds, and the A/D converter sends a number to the computer each time the interrupt arrives. An alternative implementation is simply to access the A/D upon completion of each cycle of the code execution, a scheme often referred to as **free running**. A further alternative is to use some other device to determine a sample, such as an encoder on an engine crankshaft that supplies a pulse to trigger a computer cycle. This scheme is referred to as **event-based** sampling. In the first case the sample period is precisely fixed; in the second case the sample period is essentially fixed by the length of the code, providing no logic branches are present that could vary the amount of code executed; in the third case, the sample period varies with the engine speed. Thus in Fig. 1.1 we identify the sequence of numbers into the computer as  $\hat{e}(kT)$ . We conclude from the periodic sampling action of the A/D converter that some of the signals in the digital control system, like  $\hat{e}(kT)$ , are variable only at discrete times. We call these variables **discrete signals** to distinguish them from variables like  $w$  and  $y$ , which change continuously in time. A system having both discrete and continuous signals is called a **sampled-data** system.

quantization

In addition to generating a discrete signal, however, the A/D converter also provides a **quantized** signal. By this we mean that the output of the A/D converter must be stored in digital logic composed of a finite number of digits. Most commonly, of course, the logic is based on binary digits (i.e., bits) composed

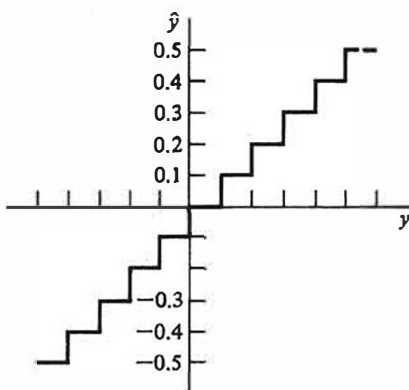
<sup>1</sup> See especially the book by Bode (1945).

of 0's and 1's, but the essential feature is that the representation has a finite number of digits. A common situation is that the conversion of  $y$  to  $\hat{y}$  is done so that  $\hat{y}$  can be thought of as a number with a fixed number of places of accuracy. If we plot the values of  $y$  versus the resulting values of  $\hat{y}$  we can obtain a plot like that shown in Fig. 1.2. We would say that  $\hat{y}$  has been truncated to one decimal place, or that  $\hat{y}$  is *quantized* with a  $q$  of 0.1, since  $\hat{y}$  changes only in fixed quanta of, in this case, 0.1 units. (We will use  $q$  for quantum size, in general.) Note that quantization is a nonlinear function. A signal that is both discrete and quantized is called a **digital signal**. Not surprisingly, digital computers in this book process digital signals.

emulation

In a real sense the problems of analysis and design of *digital controls* are concerned with taking account of the effects of the sampling period  $T$  and the quantization size  $q$ . If both  $T$  and  $q$  are extremely small (sampling frequency 30 or more times the system bandwidth with a 16-bit word size), digital signals are nearly continuous, and continuous methods of analysis and design can be used. The resulting design could then be converted to the digital format for implementation in a computer by using the simple methods described in Chapter 3 or the **emulation** method described in Chapter 7. We will be interested in this text in gaining an understanding of the effects of all sample rates, fast and slow, and the effects of quantization for large and small word sizes. Many systems are originally conceived with fast sample rates, and the computer is specified and frozen early in the design cycle; however, as the designs evolve, more demands are placed on the system, and the only way to accommodate the increased computer load is to slow down the sample rate. Furthermore, for cost-sensitive digital systems, the best design is the one with the lowest cost computer that will do the required job. That translates into being the computer with the slowest speed and the smallest word size. We will, however, treat the problems of varying  $T$  and  $q$  separately. We first consider  $q$  to be zero and study discrete and sampled-data (combined discrete and continuous) systems that are linear. In Chapter 10 we will analyze

**Figure 1.2**  
Plot of output versus  
input characteristics of  
the A/D converter



in more detail the source and the effects of quantization, and we will discuss in Chapters 7 and 11 specific effects of sample-rate selection.

Our approach to the design of digital controls is to assume a background in continuous systems and to relate the comparable digital problem to its continuous counterpart. We will develop the essential results, from the beginning, in the domain of discrete systems, but we will call upon previous experience in continuous-system analysis and in design to give alternative viewpoints and deeper understanding of the results. In order to make meaningful these references to a background in continuous-system design, we will review the concepts and define our notation in Chapter 2.

## 1.2 Overview of Design Approach

An overview of the path we plan to take toward the design of digital controls will be useful before we begin the specific details. As mentioned above, we place systems of interest in three categories according to the nature of the signals present. These are discrete systems, sampled-data systems, and digital systems.

In discrete systems all signals vary at discrete times only. We will analyze these in Chapter 4 and develop the  $z$ -transform of discrete signals and “pulse”-transfer functions for linear constant discrete systems. We also develop discrete transfer functions of continuous systems that are sampled, systems that are called sampled-data systems. We develop the equations and give examples using both transform methods and state-space descriptions. Having the discrete transfer functions, we consider the issue of the dynamic response of discrete systems.

A sampled-data system has both discrete and continuous signals, and often it is important to be able to compute the continuous time response. For example, with a slow sampling rate, there can be significant **ripple** between sample instants. Such situations are studied in Chapter 5. Here we are concerned with the question of data extrapolation to convert discrete signals as they might emerge from a digital computer into the continuous signals necessary for providing the input to one of the plants described above. This action typically occurs in conjunction with the D/A conversion. In addition to data extrapolation, we consider the analysis of sampled signals from the viewpoint of continuous analysis. For this purpose we introduce impulse modulation as a model of sampling, and we use Fourier analysis to give a clear picture for the ambiguity that can arise between continuous and discrete signals, also known as **aliasing**. The plain fact is that more than one continuous signal can result in exactly the same sample values. If a sinusoidal signal,  $y_1$ , at frequency  $f_1$  has the same samples as a sinusoid  $y_2$  of a *different* frequency  $f_2$ ,  $y_1$  is said to be an **alias** of  $y_2$ . A corollary of aliasing is the **sampling theorem**, which specifies the conditions necessary if this ambiguity is to be removed and only one continuous signal allowed to correspond to a given set of samples.

aliasing

## digital filters

As a special case of discrete systems and as the basis for the emulation design method, we consider discrete equivalents to continuous systems, which is one aspect of the field of **digital filters**. Digital filters are discrete systems designed to process discrete signals in such a fashion that the digital device (a digital computer, for example) can be used to replace a continuous filter. Our treatment in Chapter 6 will concentrate on the use of discrete filtering techniques to find discrete equivalents of continuous-control compensator transfer functions. Again, both transform methods and state-space methods are developed to help understanding and computation of particular cases of interest.

## modern control

Once we have developed the tools of analysis for discrete and sampled systems we can begin the design of feedback controls. Here we divide our techniques into two categories: **transform**<sup>2</sup> and **state-space**<sup>3</sup> methods. In Chapter 7 we study the transform methods of the root locus and the frequency response as they can be used to design digital control systems. The use of state-space techniques for design is introduced in Chapter 8. For purposes of understanding the design method, we rely mainly on **pole placement**, a scheme for forcing the closed-loop poles to be in desirable locations. We discuss the selection of the desired pole locations and point out the advantages of using the optimal control methods covered in Chapter 9. Chapter 8 includes control design using feedback of all the “state variables” as well as methods for estimating the state variables that do not have sensors directly on them. In Chapter 9 the topic of **optimal control** is introduced, with emphasis on the steady-state solution for linear constant discrete systems with quadratic loss functions. The results are a valuable part of the designer’s repertoire and are the only techniques presented here suitable for handling multivariable designs. A study of quantization effects in Chapter 10 introduces the idea of random signals in order to describe a method for treating the “average” effects of this important nonlinearity.

## identification

The last four chapters cover more advanced topics that are essential for most complete designs. The first of these topics is sample rate selection, contained in Chapter 11. In our earlier analysis we develop methods for examining the effects of different sample rates, but in this chapter we consider for the first time the question of sample rate as a design parameter. In Chapter 12, we introduce **system identification**. Here the matter of model making is extended to the use of experimental data to verify and correct a theoretical model or to supply a dynamic description based only on input–output data. Only the most elementary of the concepts in this enormous field can be covered, of course. We present the method of least squares and some of the concepts of maximum likelihood.

In Chapter 13, an introduction to the most important issues and techniques for the analysis and design of nonlinear sampled-data systems is given. The

<sup>2</sup> Named because they use the Laplace or Fourier transform to represent systems.

<sup>3</sup> The state space is an extension of the space of displacement and velocity used in physics. Much that is called **modern control theory** uses differential equations in state-space form. We introduce this representation in Chapter 4 and use it extensively afterwards, especially in Chapters 8 and 9.

analysis methods treated are the describing function, equivalent linearization, and Lyapunov's second method of stability analysis. Design techniques described are the use of inverse nonlinearity, optimal control (especially time-optimal control), and adaptive control. Chapter 14 includes a case study of a disk-drive design, and treatment of both implementation and manufacturing issues is discussed.

### 1.3 Computer-Aided Design

As with any engineering design method, design of control systems requires many computations that are greatly facilitated by a good library of well-documented computer programs. In designing practical digital control systems, and especially in iterating through the methods many times to meet essential specifications, an interactive computer-aided control system design (CACSD) package with simple access to plotting graphics is crucial. Many commercial control system CACSD packages are available which satisfy that need, MATLAB® being the most popular one. Much of the discussion in the book assumes that a designer has access to one of the CACSD products. Specific MATLAB routines that can be used for performing calculations are indicated throughout the text and in some cases the full MATLAB command sequence is shown. All the graphical figures were developed using MATLAB and the files that created them are contained in the Digital Control Toolbox which is available on the Web at no charge. Files based on MATLAB with Control System Toolbox are available at [elliskaglepress.com](http://elliskaglepress.com). These figure files should be helpful in understanding the specifics on how to do a calculation and are an important augmentation to the book's examples.

CACSD support for a designer is now universal; however, it is essential that the designer is able to work out very simple problems by hand in order to have some idea about the reasonableness of the computer's answers. Having the knowledge of doing the calculations by hand is also critical for identifying trends that guide the designer; the computer can identify problems but the designer must make intelligent choices in guiding the refinement of the computer design.

### 1.4 Suggestions for Further Reading

Several histories of feedback control are readily available, including a *Scientific American Book* (1955), and the study of Mayr (1970). A good discussion of the historical developments of control is given by Dorf (1980) and by Fortmann and Hitz (1977), and many other references are cited by these authors for the

MATLAB

Digital Control Toolbox

interested reader. One of the earliest published studies of control systems operating on discrete time data (sampled-data systems in our terminology) is given by Hurewicz in Chapter 5 of the book by James, Nichols, and Phillips (1947). Another is by Ragazzini and Franklin (1958).

The ideas of tracking and robustness embody many elements of the objectives of control system design. The concept of tracking contains the requirements of system stability, good transient response, and good steady-state accuracy, all concepts fundamental to every control system. Robustness is a property essential to good performance in practical designs because real parameters are subject to change and because external, unwanted signals invade every system. Discussion of performance specifications of control systems is given in most books on introductory control, including Franklin, Powell, and Emami-Naeini (2019). We will study these matters in later chapters with particular reference to digital control design.

To obtain a firm understanding of dynamics, we suggest a comprehensive text by Cannon (1967) or chapter 2 in Franklin, Powell, and Emami-Naeini (2019). They are concerned with writing the equations of motion of physical systems in a form suitable for control studies.

## 1.5 Summary

- In a digital control system, the analog electronics used for compensation in a continuous system is replaced with a digital computer or microcontroller, an analog-to-digital (A/D) converter, and a digital-to-analog (D/A) converter.
- Design of a digital control system can be accomplished by transforming a continuous design, called emulation, or designing the digital system directly. Either method can be carried out using transform or state-space system description.
- The design of a digital control system includes determining the effect of the sample rate and selecting a rate that is sufficiently fast to meet all specifications.
- Most designs today are carried out using computer-based methods; however the designer needs to know the hand-based methods in order to intelligently guide the computer design as well as to have a sanity check on its results.

## 1.6 Problems

- 1.1 Suppose a radar search antenna at the San Francisco airport rotates at 6 rev/min, and data points corresponding to the position of flight 1081 are plotted on the controller's screen once per antenna revolution. Flight 1081 is traveling directly toward the airport at 540 mi/hr. A feedback control system is established through the controller who gives course corrections to the pilot. He wishes to do so each 9 mi of travel of the aircraft, and his instructions consist of course headings in integral degree values.

- (a) What is the sampling rate, in Hz, of the range signal plotted on the radar screen?
  - (b) What is the sampling rate, in Hz, of the controller's instructions?
  - (c) Identify the following signals as continuous, discrete, or digital:
    - i. the aircraft's range from the airport,
    - ii. the range data as plotted on the radar screen,
    - iii. the controller's instructions to the pilot,
    - iv. the pilot's actions on the aircraft control surfaces.
  - (d) Is this a continuous, sampled-data, or digital control system?
  - (e) Show that it is possible for the pilot of flight 1081 to fly a zigzag course which would show up as a straight line on the controller's screen. What is the (lowest) frequency of a sinusoidal zigzag course which will be hidden from the controller's radar?
- 1.2 If a signal varies between 0 and 10 volts (called the **dynamic range**) and it is required that the signal must be represented in the digital computer to the nearest 5 millivolts, that is, if the *resolution* must be 5 mv, determine how many bits the analog-to-digital converter must have.
  - 1.3 Describe five digital control systems that you are familiar with. State what you think the advantages of the digital implementation are over an analog implementation.
  - 1.4 Historically, house heating system thermostats were a bimetallic strip that would make or break the contact depending on temperature. Today, most thermostats are digital. Describe how you think they work and list some of their benefits.
  - 1.5 Use MATLAB (obtain a copy of the Student Edition or use what's available to you) and plot  $y$  vs  $x$  for  $x = 1$  to 10 where  $y = x^2$ . Label each axis and put a title on it.
  - 1.6 Use MATLAB (obtain a copy of the Student Edition or use what's available to you) and make two plots (use MATLAB's subplot) of  $y$  vs  $x$  for  $x = 1$  to 10. Put a plot of  $y = x^2$  on the top of the page and  $y = \sqrt{x}$  on the bottom.



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## Review of Continuous Control

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### A Perspective on the Review of Continuous Control

The purpose of this chapter is to provide a ready reference source of the material that you have already taken in a prerequisite course. The presentation is not sufficient to learn the material for the first time; rather, it is designed to state concisely the key relationships for your reference as you move to the new material in the ensuing chapters. For a more in-depth treatment of any of the topics, see an introductory control text such as *Feedback Control of Dynamic Systems*, by Franklin, Powell, and Emami-Naeini (2019).

### Chapter Overview

The chapter reviews the topics normally covered in an introductory controls course: dynamic response, feedback properties, root-locus design, frequency response design, and state-space design.

## 2.1 Dynamic Response

In control system design, it is important to be able to predict how well a trial design matches the desired performance. We do this by analyzing the equations of the system model. The equations can be solved using linear analysis approximations or simulated via numerical methods. Linear analysis allows the designer to examine quickly many candidate solutions in the course of design iterations and is, therefore, a valuable tool. Numerical simulation allows the designer to check the final design more precisely including all known characteristics and is discussed in Section 13.2. The discussion below focuses on linear analysis.

### 2.1.1 Differential Equations

state-variable form

Linear dynamic systems can be described by their differential equations. Many systems involve coupling between one part of a system and another. Any set of differential equations of any order can be transformed into a coupled set of first-order equations called the **state-variable form**. So a general way of expressing the dynamics of a linear system is

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}u \quad (2.1)$$

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{J}u. \quad (2.2)$$

where the column vector  $\mathbf{x}$  is called the **state** of the system and contains  $n$  elements for an  $n$ th-order system,  $u$  is the  $m \times 1$  input vector to the system,  $\mathbf{y}$  is the  $p \times 1$  output vector,  $\mathbf{F}$  is an  $n \times n$  system matrix,  $\mathbf{G}$  is an  $n \times m$  input matrix,  $\mathbf{H}$  is a  $p \times n$  output matrix, and  $\mathbf{J}$  is  $p \times m$ .<sup>1</sup> Until Chapter 9, all systems will have a scalar input,  $u$ , and a scalar output  $y$ ; in this case,  $\mathbf{G}$  is  $n \times 1$ ,  $\mathbf{H}$  is  $1 \times n$ , and  $\mathbf{J}$  is a scalar.

Using this system description, we see that the second-order differential equation

$$\ddot{y} + 2\zeta\omega_o\dot{y} + \omega_o^2 y = K_o u, \quad (2.3)$$

can be written in the state-variable form as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_o^2 & -2\zeta\omega_o \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ K_o \end{bmatrix} u \quad (2.4)$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

state

where the state

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$

is the vector of variables necessary to describe the future behavior of the system, given the initial conditions of those variables.

### 2.1.2 Laplace Transforms and Transfer Functions

The analysis of linear systems is facilitated by use of the Laplace transform. The most important property of the Laplace transform (with zero initial conditions) is the transform of the derivative of a signal.

$$\mathcal{L}\{\dot{f}(t)\} = sF(s). \quad (2.5)$$

<sup>1</sup> It is also common to use  $\mathbf{A}$ ,  $\mathbf{B}$ ,  $\mathbf{C}$ ,  $\mathbf{D}$  in place of  $\mathbf{F}$ ,  $\mathbf{G}$ ,  $\mathbf{H}$ ,  $\mathbf{J}$  as MATLAB does throughout. We prefer to use  $\mathbf{F}$ ,  $\mathbf{G}$  ... for a continuous plant description,  $\mathbf{A}$ ,  $\mathbf{B}$  ... for compensation, and  $\Phi$ ,  $\Gamma$  ... for the discrete plant description in order to delineate the various system equation usages.

This relation enables us to find easily the transfer function,  $G(s)$ , of a linear continuous system, given the differential equation of that system. So we see that Eq. (2.3) has the transform

$$(s^2 + 2\zeta\omega_o s + \omega_o^2)Y(s) = K_o U(s),$$

and, therefore, the transfer function,  $G(s)$ , is

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K_o}{s^2 + 2\zeta\omega_o s + \omega_o^2}.$$

CACSD software typically accepts the specification of a system in either the state-variable form or the transfer function form. The quantities specifying the state-variable form (Eqs. 2.1 and 2.2) are  $\mathbf{F}$ ,  $\mathbf{G}$ ,  $\mathbf{H}$ , and  $\mathbf{J}$ . This is referred to as the "ss" form in MATLAB. The transfer function is specified in a polynomial form ("tf") or a factored zero-pole-gain form ("zpk"). The transfer function in polynomial form is

$$G(s) = \frac{b_1 s^m + b_2 s^{m-1} + \cdots + b_{m+1}}{a_1 s^n + a_2 s^{n-1} + \cdots + a_{n+1}}, \quad (2.6)$$

where the MATLAB quantity specifying the numerator is a  $1 \times (m+1)$  matrix of the coefficients, for example

$$\text{num} = [b_1 \quad b_2 \quad \cdots \quad b_{m+1}]$$

and the quantity specifying the denominator is a  $1 \times (n+1)$  matrix, for example

$$\text{den} = [a_1 \quad a_2 \quad \cdots \quad a_{n+1}].$$

In MATLAB with Control System Toolbox the numerator and denominator are combined into one system specification with the statement

$$\text{sys} = \text{tf}(\text{num}, \text{den}).$$

In the zero-pole-gain form, the transfer function is written as the ratio of two factored polynomials,

$$G(s) = K \frac{\prod_{i=1}^m (s - z_i)}{\prod_{i=1}^n (s - p_i)}, \quad (2.7)$$

and the quantities specifying the transfer function are an  $m \times 1$  matrix of the zeros, an  $n \times 1$  matrix of the poles, and a scalar gain, for example

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \cdots \\ z_m \end{bmatrix}, \quad p = \begin{bmatrix} p_1 \\ p_2 \\ \cdots \\ p_n \end{bmatrix}, \quad k = K$$

and can be combined into a system description by

$$\text{sys} = \text{zpk}(z,p,k).$$

For the equations of motion of a system with second-order or higher equations, the easiest way to find the transfer function is to use Eq. (2.5) and do the math by hand. If the equations of motion are in the state-variable form and the transfer function is desired, the Laplace transform of Eqs. (2.1) and (2.2) yields

$$G(s) = \frac{y(s)}{u(s)} = \mathbf{H}(s\mathbf{I} - \mathbf{F})^{-1}\mathbf{G} + J.$$

In MATLAB, given  $\mathbf{F}$ ,  $\mathbf{G}$ ,  $\mathbf{H}$ , and  $J$ , one can find the polynomial transfer function form by the MATLAB script

$$\text{sys} = \text{tf}(\text{ss}(\mathbf{F},\mathbf{G},\mathbf{H},J))$$

or the zero-pole-gain form by

$$\text{sys} = \text{zpk}(\text{ss}(\mathbf{F},\mathbf{G},\mathbf{H},J)).$$

Likewise, one can find a state-space realization of a transfer function by

$$\text{sys} = \text{ss}(\text{tf}(\text{num},\text{den})).$$

### 2.1.3 Output Time Histories

Given the transfer function and the input,  $u(t)$ , with the transform  $U(s)$ , the output is the product,

$$Y(s) = G(s)U(s). \quad (2.8)$$

The transform of a time function can be found by use of a table (See Appendix B.2); however, typical inputs considered in control system design are steps

$$u(t) = R_o 1(t), \quad \Rightarrow \quad U(s) = \frac{R_o}{s},$$

ramps

$$u(t) = V_o t 1(t), \quad \Rightarrow \quad U(s) = \frac{V_o}{s^2},$$

parabolas

$$u(t) = \frac{A_o t^2}{2} 1(t), \quad \Rightarrow \quad U(s) = \frac{A_o}{s^3},$$

and sinusoids

$$u(t) = B \sin(\omega t) 1(t), \quad \Rightarrow \quad U(s) = \frac{B\omega}{s^2 + \omega^2}.$$

Using Laplace transforms, the output  $Y(s)$  from Eq. (2.8) is expanded into its elementary terms using partial fraction expansion, then the time function associated with each term is found by looking it up in the table. The total time function,  $y(t)$ , is the sum of these terms. In order to do the partial fraction expansion, it is necessary to factor the denominator. Typically, only the simplest cases are analyzed this way. Usually, system output time histories are solved numerically using computer based methods such as MATLAB's `step.m` for a step input or `lsim.m` for an arbitrary input time history. However, useful information about system behavior can be obtained by finding the individual factors without ever solving for the time history, a topic to be discussed later. These will be important because specifications for a control system are frequently given in terms of these time responses.

### 2.1.4 The Final Value Theorem

A key theorem involving the Laplace transform that is often used in control system analysis is the **final value theorem**. It states that, if the system is stable and has a final, constant value

$$\lim_{t \rightarrow \infty} x(t) = x_{ss} = \lim_{s \rightarrow 0} sX(s). \quad (2.9)$$

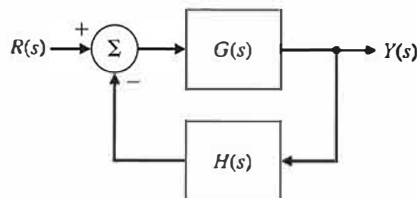
The theorem allows us to solve for that final value without solving for the system's entire response. This will be very useful when examining steady-state errors of control systems.

### 2.1.5 Block Diagrams

Manipulating block diagrams is useful in the study of feedback control systems. The most common and useful result is that the transfer function of the feedback system shown in Fig. 2.1 reduces to

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + H(s)G(s)}. \quad (2.10)$$

**Figure 2.1**  
An elementary feedback system



### 2.1.6 Response versus Pole Locations

Given the transfer function of a linear system,

$$H(s) = \frac{b(s)}{a(s)},$$

poles

zeros

impulse response

the values of  $s$  such that  $a(s) = 0$  will be places where  $H(s)$  is infinity, and these values of  $s$  are called **poles** of  $H(s)$ . On the other hand, values of  $s$  such that  $b(s) = 0$  are places where  $H(s)$  is zero, and the corresponding  $s$  locations are called **zeros**. Since the Laplace transform of an impulse is unity, the **impulse response** is given by the time function corresponding to the transfer function. Each pole location in the  $s$ -plane can be identified with a particular type of response. In other words, the poles identify the classes of signals contained in the impulse response, as may be seen by a partial fraction expansion of  $H(s)$ . For a first order pole

$$H(s) = \frac{1}{s + \sigma},$$

Table B.2, Entry 8, indicates that the impulse response will be an exponential function; that is

$$h(t) = e^{-\sigma t} 1(t).$$

stability

time constant

When  $\sigma > 0$ , the pole is located at  $s < 0$ , the exponential decays, and the system is said to be **stable**. Likewise, if  $\sigma < 0$ , the pole is to the right of the origin, the exponential grows with time and is referred to as **unstable**. Figure 2.2 shows a typical response and the **time constant**

$$\tau = \frac{1}{\sigma} \quad (2.11)$$

as the time when the response is  $1/e$  times the initial value.

Complex poles can be described in terms of their real and imaginary parts, traditionally referred to as

$$s = -\sigma \pm j\omega_d.$$

This means that a pole has a negative real part if  $\sigma$  is positive. Since complex poles always come in complex conjugate pairs for real polynomials, the denominator corresponding to a complex pair will be

$$a(s) = (s + \sigma - j\omega_d)(s + \sigma + j\omega_d) = (s + \sigma)^2 + \omega_d^2. \quad (2.12)$$

When finding the transfer function from differential equations, we typically write the result in the polynomial form

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \quad (2.13)$$